

10 (a) Glycolysis is a stage of respiration that occurs in the cytoplasm of eukaryotic cells.

Outline the glycolysis stage of respiration.

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..... [4]

(b) A group of drugs called statins are taken by many people to help lower the concentration of cholesterol in the blood. Sometimes statins can inhibit the production of coenzyme A.

Suggest the effects of a low concentration of coenzyme A on aerobic respiration.

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(c) Figure 10.1 shows part of a mitochondrion.

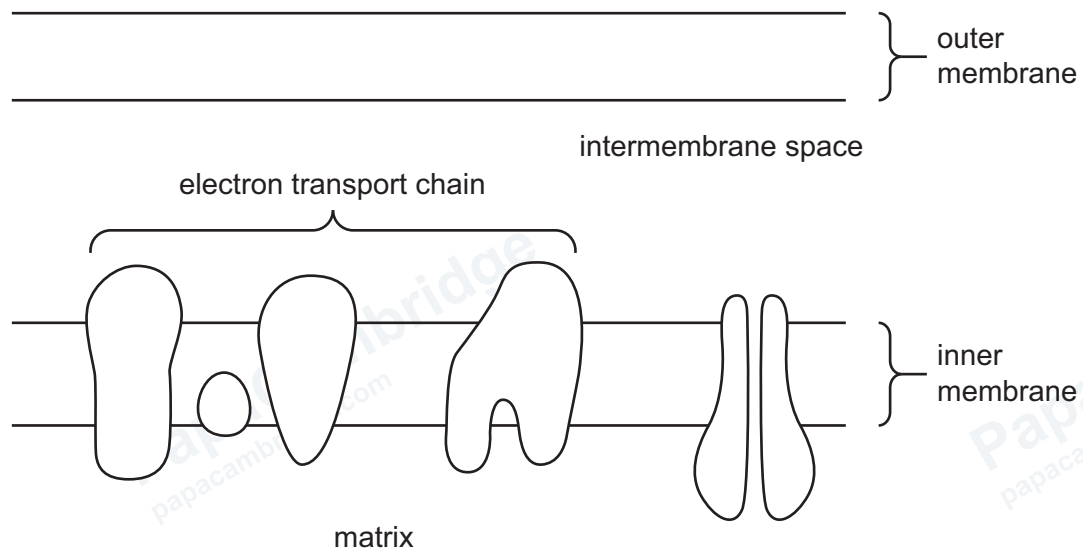


Figure 10.1

On Figure 10.1, use label lines and letters to label:

- A** – a structure that synthesises ATP
- B** – a structure that is impermeable to ions
- C** – an area that contains a high concentration of protons
- D** – a structure that receives hydrogen atoms from reduced NAD.

[4]

[Total: 12]